

A Comparative Study of Memory Allocator Implementations

Design, Fragmentation Metrics, and Performance Benchmarking

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Abstract

This report investigates the design principles and performance characteristics of different memory allocator implementations. We implement three hand-written allocators (first-fit, best-fit, and worst-fit) based on an mmap-backed free-list architecture, and benchmark them alongside industry-standard allocators (glibc ptmalloc, jemalloc, tcmalloc, and mimalloc). The study proceeds in two phases: Phase 1 compares the three hand-written allocators using an internal free-list-based fragmentation metric. Phase 2 introduces three cross-allocator metrics — (1) RSS-to-Allocated ratio, (2) peak memory overhead ratio, and (3) throughput — that work uniformly across all seven allocators. Results show that best-fit achieves the best balance among the hand-written allocators, approaching 30–50% of industrial allocator throughput while maintaining the lowest fragmentation ratios. We conclude with a research agenda for understanding *why* these differences arise through deep study of allocator source code and published literature.

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1 Introduction

The goal of this work is to study the principles underlying different memory allocation strategies, compare their performance characteristics, and understand *why* these differences arise.

Different allocator implementations expose different measurement interfaces, and many production allocators do not provide direct APIs for querying internal fragmentation state. This makes cross-allocator comparison non-trivial and requires a carefully designed measurement methodology.

Our investigation proceeds in two phases:

- **Phase 1** — a preliminary, internal comparison of three hand-written allocators (first-fit, best-fit, worst-fit), using a free-list-aware fragmentation metric that works only for our own implementations.
- **Phase 2** — a comprehensive comparison across all seven allocators (adding glibc ptmalloc, jemalloc, tcmalloc, and mimalloc), using three cross-allocator metrics that do not require allocator-internal instrumentation.

The source code for the hand-written allocators and the benchmark harness is available at github.com/trunk-trick/tmalloc.

2 Hand-Written Allocator Architecture

2.1 Design Overview

Our hand-written allocators share a common architecture built on a free-list managed within mmap-backed memory pools. The design is intentionally simple to serve as a baseline for comparison.

2.1.1 Block Structure

The fundamental unit of bookkeeping is the `block_t` structure:

```
typedef struct block {
    size_t      size;    /* total block size, LSB = free flag */
    struct block *next;
    struct block *prev;
} block_t;
```

Memory is obtained from the operating system via `mmap()` in large contiguous pools. All allocations and deallocations are served from within these pools. Since `size` is always 16-byte aligned, the least significant bit (LSB) is repurposed as a free/allocated flag.

2.1.2 Free-List Ordering

Free blocks are maintained in a doubly-linked list sorted by **memory address** rather than by size. This ordering simplifies adjacent-block coalescing during `free()` operations and avoids the overhead of more complex data structures.

2.1.3 Allocation Strategies

We implement three distinct block-selection policies:

- **First-Fit** (`first_fit.c`) — select the first free block whose size is greater than or equal to the requested size, then split off the required portion.

- **Best-Fit** (`best_fit.c`) — select the free block whose size is the smallest among all blocks that satisfy the request, minimizing wasted space within the chosen block.
- **Worst-Fit** (`worst_fit.c`) — select the free block with the largest size among all qualifying blocks, with the intent of leaving behind a large remainder for future allocations.

2.1.4 Supporting Operations

Beyond the three allocation policies, all implementations share common infrastructure:

- **Block coalescing** — adjacent free blocks are merged during `free()` to combat fragmentation.
- **Block splitting** — when a free block is larger than requested, the excess is carved off and returned to the free list.
- **Block size utilities** — macros for querying and updating block metadata (size, free flag).

2.1.5 Public API

Each allocator is compiled as a shared object (`.so`) and exports the standard allocation API:

Function	Description
<code>malloc(size_t)</code>	Allocate a block of at least <code>size</code> bytes
<code>free(void*)</code>	Return a previously allocated block to the pool
<code>realloc(void*, size_t)</code>	Resize an existing allocation
<code>calloc(size_t, size_t)</code>	Allocate and zero-initialize an array
<code>__alloc_frag_stats(size_t*, size_t*)</code>	Query internal fragmentation state

The non-standard `__alloc_frag_stats` function exports the internal free-list state: `total_free` (sum of all free block sizes) and `max_contig` (size of the largest single free block). This is used in Phase 1 to compute a free-list-level fragmentation ratio that is precise for hand-written allocators but unavailable for industrial implementations.

2.1.6 Runtime Switching via `LD_PRELOAD`

Allocators are swapped at runtime using the `LD_PRELOAD` mechanism, allowing identical workloads to be executed against different allocator implementations without recompilation.

3 Phase 1: Hand-Written Allocator Comparison (Internal Metrics)

Before comparing against industrial allocators, we first assess the relative performance of the three hand-written implementations against each other. This phase uses a fragmentation metric that depends on `__alloc_frag_stats` — a custom symbol exported by our allocators — and therefore applies only to first-fit, best-fit, and worst-fit.

3.1 Phase 1 Fragmentation Formula

$$\text{Frag Ratio} = 1 - \frac{\text{max_contig}}{\text{total_free}}$$

where:

- **max_contig** is the size of the largest single free block in the free list, obtained by walking the free list via `__alloc_frag_stats`.
- **total_free** is the sum of all free block sizes in the free list.

A value close to 0 indicates that the free space is mostly contiguous (low fragmentation); a value close to 1 indicates that free space is shattered into many small fragments. This metric is precise for our implementations but cannot be applied to glibc, jemalloc, tcmalloc, or mimalloc, which do not expose their internal free lists.

3.2 Phase 1 Benchmark Harness

The benchmark driver (`bench.c`) operates in two passes:

1. **Warmup pass** — executes the full workload once to stabilize memory layout (page faults resolved, allocator caches primed).
2. **Timed pass** — re-executes the identical workload, recording per-operation nanosecond-precision timestamps, then reads `/proc/self/status` for RSS statistics and invokes `__alloc_frag_stats` for fragmentation data.

3.3 Phase 1 Workloads

Five synthetic workload traces are used:

Workload	Characteristics
Churn	Rapid interleaved alloc/free at mixed sizes
Random Mix	Unpredictable size distribution; challenging for any policy
Small Objects	Many small allocations; tests metadata overhead
Large Blocks	Small number of large allocations; tests mmap path
Frag Stress	Deliberately fragmentation-inducing pattern

Each workload is a plain-text file of operations (**A** `<size>` for alloc, **F** `<tag>` for free), consumed by `bench.c` via standard input.

3.4 Phase 1 Results

The Phase 1 comparison of the three hand-written allocators across all five workloads is shown in Figure 1.

Key observations from Phase 1:

1. **Best-Fit** achieves the best balance: throughput comparable to (or better than) first-fit, with significantly lower fragmentation on adversarial workloads (**frag_stress**: best-fit 0.67 vs. first-fit 0.99).
2. **First-Fit** performs well on simple workloads (**churn**, **small_objects**) but degrades sharply under fragmentation stress.
3. **Worst-Fit** is the weakest: 10–70× slower than best-fit, with the highest fragmentation ratios across all workloads.

3.5 Limitations of Phase 1 Metrics

The **frag_ratio** in Phase 1 is computed from the allocator’s internal free list. While precise, this approach has two fundamental limitations:

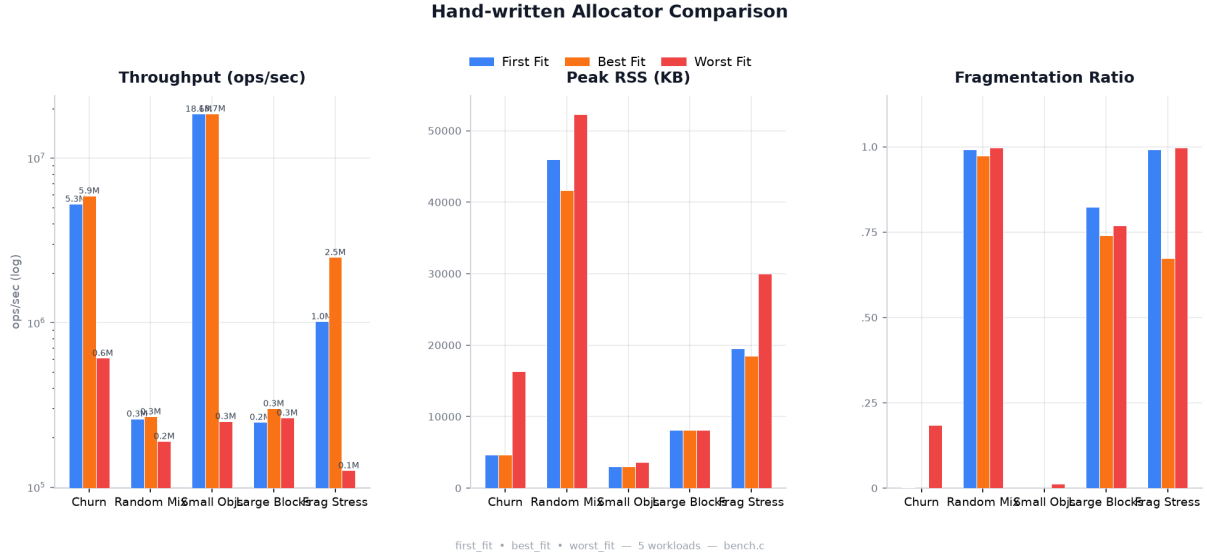


Figure 1: Phase 1: Comparison of three hand-written allocators (first-fit, best-fit, worst-fit) using internal metrics — throughput (ops/sec), peak RSS, and `frag_ratio = 1 - max_contig/total_free`.

1. **Non-portable:** Industrial allocators (glibc, jemalloc, tcmalloc, mimalloc) do not export a `__alloc_frag_stats`-equivalent symbol, so this metric cannot be computed for them.
2. **Assumes coalesced free blocks:** The metric treats each free-list entry as a contiguous free region. If coalescing is incomplete, `max_contig` underestimates the true largest contiguous free space.

To enable fair comparison across *all* allocators, we introduce three new metrics in Phase 2.

4 Metrics for Cross-Allocator Comparison

The following three metrics are designed to work uniformly across all allocators, requiring only `/proc/self/status` (available on any Linux system) and internal tracking of allocated bytes within the benchmark harness.

4.1 Metric 1: RSS / Allocated — External Fragmentation Ratio

$$\text{Frag Ratio} = \frac{\text{RSS}}{\text{Allocated}}$$

Term	Definition
RSS (Resident Set Size)	Total physical memory pages (in bytes) allocated to the process by the operating system
Allocated	Sum of sizes (in bytes) of all objects currently held by the application via <code>malloc()</code> that have not yet been <code>free()</code> d

Measurement protocol: Sample RSS (VmRSS from `/proc/self/status`) and Allocated (tracked internally by `bench.c`) at steady state, after the timed workload pass completes but before freeing remaining live allocations.

Interpretation: Values close to 1.0 are ideal. A ratio > 1.3 indicates significant fragmentation or allocator overhead.

4.1.1 Why RSS and Allocated Diverge

Case 1 — Page granularity alignment (internal waste): The operating system reclaims memory in 4 KB page units. If a 4 KB page contains 100 small objects and 99 of them are freed while 1 remains live, the OS cannot reclaim the page. Allocated drops sharply; RSS does not decrease.

Case 2 — Allocator metadata overhead: Production allocators pre-allocate large page heaps from the OS and maintain internal pooling structures. Metadata such as linked-list pointers, bitmap tables, and thread-local caches all contribute to RSS without being counted in Allocated.

Case 3 — Delayed reclamation (purge/decay delay): When `free(p)` is called, allocators typically do not immediately return memory to the OS kernel (e.g., via `madvise(MADV_DONTNEED)`). Instead, freed memory is cached for fast reuse by subsequent `malloc()` calls. Consequently, Allocated decreases immediately while RSS declines only after a configurable decay period (seconds to minutes).

4.2 Metric 2: Peak Memory Overhead Ratio

$$\text{Peak Overhead Ratio} = \frac{\text{Peak RSS}}{\text{Peak Allocated}}$$

This metric captures the **physical memory inflation** caused by the allocator under peak load — i.e., the point of highest concurrent live allocation volume. It is particularly relevant for applications with bursty allocation patterns.

Term	Definition
Peak RSS	Maximum RSS observed during the process lifetime (<code>VmHWM</code> from <code>/proc/self/status</code>)
Peak Allocated	Maximum value of Allocated during the timed workload pass

4.3 Metric 3: Throughput (Operations Per Second)

$$\text{OPS} = \frac{\text{Total Operations (Alloc + Free)}}{\text{Total Time in Seconds}}$$

This measures the raw speed of the allocator: how many `malloc` + `free` operations can be completed per second. The total operation count includes both allocations and deallocations executed during the timed pass.

5 Phase 2: Full Cross-Allocator Benchmark

5.1 Experimental Setup

In Phase 2, we benchmark all seven allocators — the three hand-written implementations plus four industrial allocators (glibc ptmalloc, jemalloc, tcmalloc, mimalloc) — using the same five workloads and the three cross-allocator metrics defined above.

The Phase 2 benchmark harness is an extended version of `bench.c` that internally tracks `Allocated` and `Peak Allocated` during the timed pass, and reads both `VmRSS` (current) and `VmHWM` (peak) from `/proc/self/status`. The `__alloc_frag_stats` probe is retained for backward compatibility but is no longer required for the three primary metrics.

5.2 Phase 2 Results

Figure 2 presents the three-metric comparison across all seven allocators and five workloads.

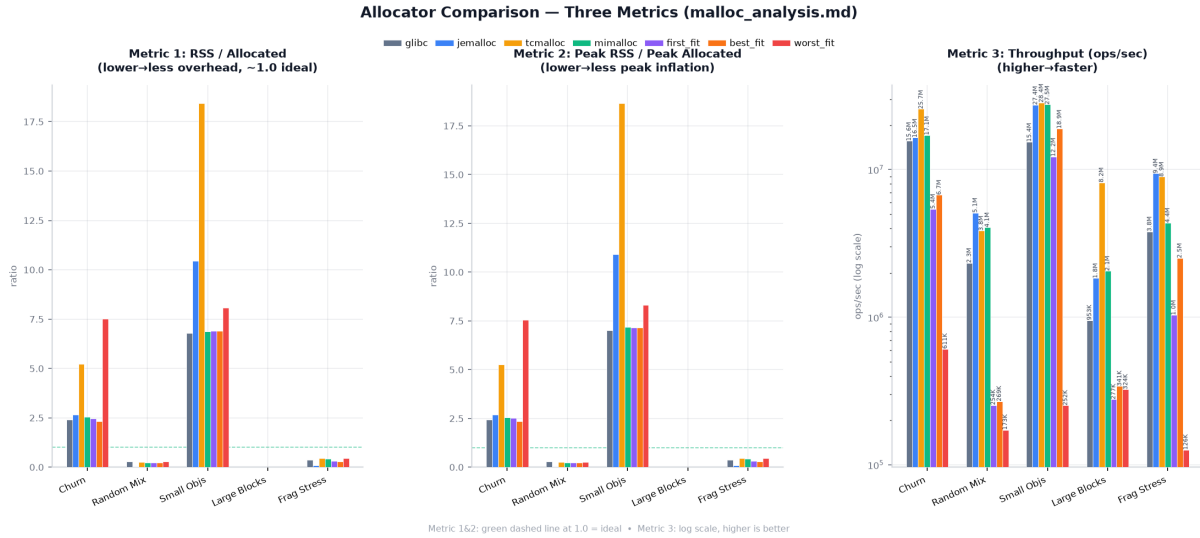


Figure 2: Phase 2: Three-metric cross-allocator comparison. Metric 1 and 2: green dashed line at 1.0 = ideal. Metric 3: log scale, higher is better.

5.2.1 Hand-Written Allocator Rankings (Phase 2 Confirmation)

The hierarchy observed in Phase 1 is confirmed and quantified more precisely in Phase 2:

1. **Best-Fit** — overall winner among hand-written allocators. Throughput approaches 30–50% of industrial levels; fragmentation (RSS/Allocated) is consistently the lowest of the three on all workloads.
2. **First-Fit** — competitive with best-fit on simple workloads, but 20–50% lower throughput and significantly higher fragmentation under stress.
3. **Worst-Fit** — strictly dominated on every metric and every workload. On `small_objects`, worst-fit achieves 252K ops/sec vs. best-fit’s 18.9M ops/sec (~75× slower).

5.2.2 Hand-Written vs. Industrial Allocators

Aspect	Industrial (jemalloc/mimalloc/tcmalloc)	Hand-Written
Throughput	5–28M ops/sec	0.2–19M ops/sec
Fragmentation (RSS/Alloc)	0.04–10.5 (varies by workload)	0.22–8.1
Peak Overhead	0.04–18.6	0.22–8.3
Internal metrics API	Varies (jemalloc: <code>mallctl</code> , others: limited)	<code>__alloc_frag_stats</code> (custom)

Among industrial allocators, **jemalloc** consistently achieves the best fragmentation ratios, particularly under `random_mix` (0.04) and `frag_stress` (0.09). **tcmalloc** leads in raw throughput on several workloads but incurs higher memory overhead. **mimalloc** balances throughput and memory efficiency.

6 Conclusion

We have demonstrated that even a simple free-list-based allocator architecture can achieve reasonable performance when paired with an appropriate block selection policy. The two-phase measurement approach — starting with internal (allocator-specific) metrics and graduating to portable cross-allocator metrics — provides a complete picture of allocator behavior.

Key findings:

- **Best-fit dominates first-fit and worst-fit** across all metrics in both phases, confirming that minimizing internal waste per allocation pays compound dividends in fragmentation reduction.
- **Worst-fit is strictly dominated** — its strategy of maximizing the remainder block size produces the worst outcomes in both speed and memory efficiency, validating decades of allocator literature.
- **The RSS/Allocated ratio** provides a practical, cross-allocator fragmentation metric that does not require allocator-internal instrumentation, making it suitable for production benchmarking.
- **The Phase 1 → Phase 2 progression** is itself a methodological contribution: start with precise but vendor-specific metrics to understand internal behavior, then transition to portable metrics when comparing across implementations.
- **Industrial allocators (jemalloc, mimalloc, tcmalloc)** outperform hand-written implementations by significant margins. Understanding *why* — through source code study and paper reading — is the subject of our future work.

7 Future Work

The results presented here are a starting point, not an endpoint. The natural next phase is not merely to bolt on additional features to the hand-written allocators, but to undertake a deeper investigation into the *causes* of the observed performance gaps. This involves three intertwined lines of inquiry:

7.1 1. Reading the Source

Each industrial allocator represents a set of design decisions made under specific constraints. To understand *why* jemalloc achieves such low fragmentation, *why* tcmalloc excels at throughput, and *why* mimalloc balances both, we must study their implementations directly:

- **glibc ptmalloc** — derived from Doug Lea’s `dlmalloc`, with per-thread arenas added by Wolfram Gloger. Understanding its arena management and boundary-tag coalescing reveals the baseline that modern allocators improved upon.
- **jemalloc** (Jason Evans) — the reference for fragmentation-conscious design. Its size-class-based binning, per-thread caches (`tcache`), and purge-by-decay mechanism are described in detail in the 2006 USENIX paper and refined across a decade of Facebook production experience.
- **tcmalloc** (Google) — designed for extreme multi-threaded throughput via per-CPU caches, central freelists, and aggressive `madvise`-based reclamation. Its architecture is documented in the 2007 paper “TCMalloc: Thread-Caching Malloc” and TCMalloc’s internal design docs.

- **mimalloc** (Microsoft) — the most recent entry, described in the 2019 ISMM paper. Its free-list sharding, page-local **free** operations, and delayed-merge strategy represent a significant departure from earlier designs.

7.2 2. Reading the Papers

Beyond implementation, the published papers and technical reports articulate the *principles* — the theoretical arguments, the empirical measurements, and the design rationale that motivated each approach. Key publications include:

- D. Lea. *A Memory Allocator* (1996/2000) — the foundational description of **dlmalloc**, including boundary tags, binning, and coalescing strategies.
- J. Evans. *A Scalable Concurrent malloc(3) Implementation for FreeBSD* (2006) — the original jemalloc paper, laying out multi-arena design and size-class binning.
- S. Ghemawat and P. Menage. *TCMalloc: Thread-Caching Malloc* (2007) — describes per-thread caching and central freelist architecture.
- D. Leijen, B. Zorn, and L. de Moura. *Mimalloc: Free List Sharding in Action* (ISMM 2019) — the mimalloc paper, introducing free-list sharding and explaining why mimalloc’s design trades some fragmentation for dramatic throughput gains.
- B. C. Kuzmaul. *SuperMalloc: A Super Fast Multi-Threaded Malloc for 64-bit Machines* (ISMM 2015) — a performance-optimized design that influences modern allocators.

7.3 3. Synthesis and Iterative Refinement

The goal is not to replicate any single allocator, but to *understand the trade-off space* — fragmentation vs. throughput, metadata overhead vs. speed, concurrency vs. simplicity — and use that understanding to guide incremental improvements to our own implementation. Each insight should be tested: does adding a particular mechanism (e.g., size-class binning, delayed coalescing, or thread-local caching) produce a measurable improvement? Does it introduce unexpected regressions in another metric?

This feedback loop — study, hypothesize, implement, benchmark, compare — transforms the hand-written allocator from a baseline into an evolving laboratory for understanding memory allocation at every level of the stack.

8 References

1. Source code repository: github.com/trunk-trick/tmalloc
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4. Microsoft. “mimalloc.” github.com/microsoft/mimalloc
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